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Machine Learning

CO2 - AT1 Analytical Problem Solving

Perception Model - student pass or fail

Given: → study hours (x_1) = 2

→ Attendance (x_2) = 1

→ weights : $w_1 = 0.6$, $w_2 = 0.4$

→ Bias = -0.5

Formula: Net Input: $w_1x_1 + w_2x_2 + b$

Calculation:-

$= (0.6 \times 2) + (0.4 \times 1) - 0.5$

$= 1.2 + 0.4 - 0.5$

$= 1.1$

step activation function:-

→ If net input ≥ 0, output = 1

→ If net input < 0, output = 0.

Since $1.1 \geq 0$, the output is 1

Conclusion: The student passes according to the perceptron model.

Backpropagation - weight update using Delta Rate.

Given:- → Initial weight = 0.5

→ Learning rate (η) = 0.1

→ Input (x) = 2

→ Actual output = 0.6

→ Target output = 1

Calculate the error

Error = Target - Actual

$= 1 - 0.6 = 0.4$

Calculate the weight update:

$\Delta w = \eta \times \text{Error} \times x$

$= 0.1 \times 0.4 \times 2$

$= 0.08$

Calculate the

$w_{new} = w_{old} + \Delta w$

$= 0.5 + 0.08$

$= 0.58$

Conclusion:- The updated weight after one iterat

3) Confusion Matrix - Disease prediction

Given:
- → True positive (TP) = 180
- → True negative (TN) = 150
- → False Positive (FP) = 20
- → False Negative (FN) = 50

Total observation:-

$N = 180 + 150 + 20 + 50$

$= 400$

1) Accuracy:-

$= \frac{TP + TN}{N} \times 100$

$= \frac{180 + 150}{400} \times 100 = 82.5\%$

2) Precision:- $= \frac{TP}{TP + FP} \times 100$

$= \frac{180}{180 + 20} \times 100$

$= 90\%$

3) Sensitivity (Recall) $= \frac{TP}{TP + FN} \times 100$

$= \frac{180}{180 + 50} \times 100$

$= 78.26\%$

4) Specificity: $\frac{TN}{TN + FP} \times 100$

$= \frac{150}{150 + 20} * 100$

$= 88.24\%$

However, its sensitivity is 78.26%, meaning some patients with the disease may be missed.

Genetic Algorithm - Maximizing

$f(x) = x^2$

Given population : 2, 4, 6, 8

Calculate fitness values

Individual (x)	Fitness (x^2)
2	4
4	16
6	36
8	64

Select the best candidates

The individuals with the highest fitness values are:-

→ x = 8 fitness = 64

→ x = 6 fitness = 36

Therefore, 6 and 8 are selected as parents.

Crossover operation:-

Represent the selected individuals in 4-bit binary :-

→ 6 = 0110

→ 8 = 1000

Choose a single crossover point after the 2nd bit :-

→ parent 1 : 01 | 10

→ parent 2 : 10 | 00

Exchange the right-side bits:-

→ offspring 1 : 0100 = 4

→ offspring 2 : 1010 = 10

Mutation randomly changes one (or) more bits in an individual.

Conclusion:- Selection chooses fitter individuals, crossover creates new offspring, and mutation helps explore better solutions.

5) Deep Neural Network - Training & validation Accuracy :-

Given :- → Training accuracy increases from 60-1. to 95-1.

→ Validation accuracy initially increases from 58-1. to ~~7~~ 1.

then decreases to 68-1.

Issue : Overfitting

Overfitting occurs when a model learns the training data too closely, including noise and unnecessary details. As a result, it performs well on training data but poorly on unseen data.

Reason :- The training accuracy continues to improve, while validation accuracy decreases. This indicate that the model is not generalizing well to new data.

Two techniques to overcome overfitting :-

1) Dropout :- Randomly disables some neurons during training to prevent excessive dependence on particular neurons.

2) Early stopping :- stops training when validation performance begins to deteriorate, preventing the model from learning unnecessary details.

Importance of validation accuracy :-

Validation accuracy measures how well the model performs on data not used to update its weights. It helps identify overfitting, compare models, and select the best training stage.

Conclusion :- The model is affected by overfitting.

Dropout and early stopping can improve generalization and help the model perform better on unseen data.